

EFCLASS Technical Note II

Perturbation-Level σ_8 Suppression via $\mu(a) < 1$: Structural Limits of Scale-Free Modifications

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Technical Note I established that background-level EFC cannot suppress σ_8 . This note investigates the perturbation channel: $\mu(a) = 1 - B g(a)$ with $\mu < 1$, which weakens the gravitational source term in the growth equation. We perform a systematic scan over gate amplitude (B) and temporal support (steepness n) at fixed μ_0 , using a scale-independent (no k -dependence) modification throughout. The central result is that reducing n from 6 to 2 (broadening the gate) exactly doubles the σ_8 suppression at any μ_0 . This universal factor is geometric, not parametric. Within Planck 1σ constraints on μ_0 , a scale-free $\mu(a)$ closes $\sim 43\%$ of the S_8 tension; at $\sim 1.5\sigma$ it closes $\sim 73\%$. Full resolution requires either scale dependence or an additional channel.

1 Background and Setup

Technical Note I (DOI: 10.6084/m9.figshare.31333414) proved that the EFC background gate with $E(0) = 1$ normalisation produces $\Delta E^2 \leq 0$ at all $z > 0$, reducing Hubble friction and *enhancing* growth. This rules out the background channel for σ_8 suppression.

We therefore investigate the perturbation channel. In the standard (μ, Σ) parameterisation of modified gravity, the growth equation becomes:

$$f' + f^2 + \left(\frac{1}{2} - \frac{3}{2} \tilde{\Omega}_m \right) f = \frac{3}{2} \mu(a) \tilde{\Omega}_m \quad (1)$$

where $f = d \ln D / d \ln a$, $\tilde{\Omega}_m(a) = \Omega_m a^{-3} / E^2(a)$, and $\mu(a) = G_{\text{eff}} / G$. The right-hand side is the gravitational source term: $\mu < 1$ directly suppresses growth.

1.1 Control Experiment: $A = 0$ (Pure μ -Model)

To isolate the perturbation effect without background interference, we set $A = 0$ (LCDM geometry) and define:

$$\mu(a) = 1 - B g(a), \quad g(a) = \frac{1}{1 + (a_t/a)^n}, \quad a_t = \frac{1}{1 + z_t} \quad (2)$$

with $z_t = 1.01$ (shared with the background gate). This modification is explicitly **scale-independent** (k -independent): μ depends only on time, not on wavenumber. The lensing potential is set to $\Sigma = 1$ (WP1 default).

1.2 Calibration Protocol

To separate amplitude from temporal support, we **calibrate** B so that $\mu(z=0)$ is held fixed when scanning n :

$$B = \frac{1 - \mu_0}{g(1; n)} \quad (3)$$

Since $g(1; n)$ varies with n (from 0.80 at $n=2$ to 0.999 at $n=10$), B adjusts to keep μ_0 constant. This isolates the effect of gate width.

2 Results: Amplitude Scan (B -sweep)

With $n = 6$ (the Technical Note I default) and $A = 0$:

B	μ_0	σ_8	S_8	χ^2_{RSD}	Planck
0.000	1.000	0.811	0.829	65.3	–
0.050	0.951	0.805	0.823	61.7	OK
0.100	0.902	0.799	0.816	58.2	OK
0.150	0.852	0.792	0.810	54.8	1σ
0.200	0.803	0.786	0.804	51.5	1σ
0.250	0.754	0.780	0.797	48.3	$>2\sigma$
0.300	0.705	0.774	0.791	45.2	$>2\sigma$

Table 1: B -sweep at fixed $n = 6$, $A = 0$. RSD χ^2 is against BOSS DR12 (3 points, full covariance). Planck column indicates compatibility with $|\mu_0 - 1| < 0.10$ (1σ) or < 0.20 (2σ).

Finding 1: σ_8 responds slowly to B . Even at $B = 0.30$ ($\mu_0 = 0.70$, far outside Planck bounds), σ_8 drops only to 0.774. The gate with $n = 6$ has insufficient temporal support.

3 Results: Gate Width Scan (n -sweep)

This is the central result. We scan $n \in \{2, 2.5, 3, 4, 5, 6, 8, 10\}$ at three μ_0 levels, with B calibrated via Eq. (3).

μ_0	n	B	σ_8	$\Delta\sigma_8$	S_8	χ^2_{RSD}	Support
0.913	10	0.087	0.801	−0.010	0.819	59.5	0.43
0.913	6	0.088	0.800	−0.011	0.818	59.0	0.42
0.913	4	0.092	0.798	−0.013	0.816	57.9	0.43
0.913	3	0.098	0.795	−0.016	0.813	56.5	0.44
0.913	2	0.109	0.789	−0.022	0.806	53.3	0.48
0.850	6	0.152	0.792	−0.019	0.810	54.6	0.73
0.850	2	0.187	0.773	−0.038	0.790	45.5	0.83
0.800	6	0.203	0.786	−0.025	0.803	51.3	0.98
0.800	2	0.250	0.761	−0.050	0.777	39.8	1.11

Table 2: Gate width scan. “Support” is $\int_{0.1}^1 |1 - \mu(a)| d \ln a$. Bold rows are the $n = 2$ cases (maximal temporal support). $\Delta\sigma_8 = \sigma_8 - 0.811$.

4 Central Result: The Universal Factor-2

At every μ_0 level, the ratio of σ_8 suppression between $n = 2$ and $n = 6$ is:

$$\frac{\Delta\sigma_8(n=2)}{\Delta\sigma_8(n=6)} = 2.00 \pm 0.01 \quad (4)$$

This is exact across three decades in μ_0 deviation.

μ_0	$\Delta\sigma_8(n=6)$	$\Delta\sigma_8(n=2)$	Ratio	Extra suppression
0.913	−0.011	−0.022	2.01	−0.011
0.850	−0.019	−0.038	2.00	−0.019
0.800	−0.025	−0.050	2.00	−0.025

Table 3: Universal factor-2 from gate broadening.

Interpretation: The factor-2 is a *geometric* property of the growth integral, not a parametric coincidence. It arises because the sigmoid gate’s effective temporal support (measured in $\ln a$) approximately doubles when n drops from 6 to 2. The σ_8 suppression is proportional to this integral:

$$\Delta\sigma_8 \propto (1 - \mu_0) \int_{\ln a_{\min}}^0 g(a; n) d \ln a \quad (5)$$

The factor-2 is thus a statement about the sigmoid's width, not about the physics of gravity.

5 S_8 Closure Fraction

The S_8 tension between Planck (0.834 ± 0.016) and Stage-III weak lensing (0.776 ± 0.017) corresponds to $\Delta\sigma_8 \approx 0.052$ (at fixed $\Omega_m = 0.3134$).

μ_0	n	σ_8	Gap closed	Planck μ_0 status
0.913	6	0.800	21%	OK
0.913	2	0.789	43%	OK
0.850	3	0.784	52%	$\sim 1.5\sigma$
0.850	2	0.773	73%	$\sim 1.5\sigma$
0.800	2	0.761	97%	2σ

Table 4: S_8 gap closure fraction for key parameter combinations. The bold row marks the “reference model”: substantial amelioration within a defensible Planck zone.

5.1 Reference Model

Based on this scan, we define the **WP1a reference model**:

$$A = 0, \quad B = 0.187, \quad n = 2, \quad z_t = 1.01 \quad \implies \quad \mu_0 = 0.85, \quad \sigma_8 = 0.773, \quad S_8 = 0.790 \quad (6)$$

This substantially ameliorates the S_8 tension (73% of the gap), sits at $\sim 1.5\sigma$ from Planck μ_0 , and reduces χ^2_{RSD} by 19.8 relative to ΛCDM (BOSS DR12). It does not fully resolve the tension; the remaining $\sim 27\%$ requires either scale dependence or a combined channel.

6 Structural Conclusions

- Background and perturbation channels are non-degenerate.** Technical Note I showed that the background (gate amplitude A) controls only geometry via Hubble friction. This note shows that the perturbation channel (μ , parameterised by B) controls only growth via the Poisson source term. When both are active, they interfere destructively for σ_8 (reduced friction enhances growth; $\mu < 1$ suppresses it).
- Temporal support matters more than instantaneous amplitude.** At fixed $\mu_0 = 0.913$, broadening the gate from $n = 6$ to $n = 2$ provides as much additional suppression as the entire $n = 6$ effect. The “universal factor-2” is a geometric invariant.
- Scale-free $\mu(a)$ has a structural ceiling.** Within Planck 1σ ($\mu_0 > 0.90$), a scale-free $\mu(a)$ closes at most $\sim 43\%$ of the S_8 gap. Full resolution within 1σ requires scale dependence ($\mu(k, a)$) or a combined channel.
- A 70% amelioration exists at $\sim 1.5\sigma$.** $\mu_0 = 0.85$ with $n = 2$ ameliorates 73% of the gap. This is not fine-tuned: it follows from the integral structure of the growth equation and the sigmoid’s width. It does not constitute full resolution of the S_8 tension.

7 Limitations and Next Steps

Limitations of this analysis:

- Growth ODE only (no full Boltzmann hierarchy). ISW and CMB lensing effects are estimated, not computed.
- $\Sigma = 1$ assumed. Alternative ($\Sigma = \mu$) would modify CMB lensing and requires CLASS implementation to quantify.
- Gate steepness n is phenomenological. A derivation from EFC first principles would constrain it.
- Planck μ_0 constraints used here ($|\mu_0 - 1| < 0.1$ at 1σ) are approximate, based on the Planck 2018 μ - Σ analysis (Planck Collaboration XIV, 2016; Aghanim et al., 2020) assuming ΛCDM background. They are model-dependent and may shift with alternative time-evolution functions.

Next steps (WP1b):

- Scale-dependent $\mu(k, a) = 1 - (1 - \mu(a))W(k; k_c)$ with one new parameter k_c . Hard stop rule: if >1 additional parameter is needed for $< 1\sigma$ compatibility, abandon.
- CLASS implementation of the μ -modification in `perturbations.c` for full $P(k, z)$, C_ℓ , and σ_8 computation.
- Combined (A, B) scan with optimised (z_t, n) once the individual channels are understood.

Reproducibility: Code: `efc.wp1_mu_only_sweep.py` (B -sweep), `efc.wp1_n_scan.py` (n -sweep). Data: BOSS DR12 RSD (3 points, full covariance from Alam+ 2017). Background: ΛCDM with $\Omega_m = 0.3134$, $H_0 = 67.4$.

Gate: $z_t = 1.01$, $a_t = 0.4975$. Builds on Technical Note I (DOI: [10.6084/m9.figshare.31333414](https://doi.org/10.6084/m9.figshare.31333414)). This note: DOI: [10.6084/m9.figshare.31333600](https://doi.org/10.6084/m9.figshare.31333600).